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Quality Tool-Agent Governance Audit Report

Quality Tool-Agent Governance Audit Report

Voyager V3 · Reliability · 6 Failure Count · L3 High Risk · Candidate

Fail

Fail → FAIL

Agent **Voyager V3**; Isolated Sandbox (Shadow Environment, isolated from production) **12** times, Success Count **6** times、Failure Count **6** times, **Reliability 50.0%**, below historical best %, alert triggered **100%**, Evaluation Gate **Fail**, Take one of five states from SanityOps Core Specification §6.2 Release Summary Status; Fail → **FAIL FAIL**。

Current Reliability reached or exceeded historical best: $R = \text{Success Count} / \text{Execution Count (n)} \times 100\% = 6 / 12 \times 100\% = 50.0\%$

$R \geq \text{Threshold} \rightarrow \text{Pass}$; $R < \text{Threshold} \rightarrow \text{Fail}$ (Quality Tool-Agent Whitepaper §4.3 Evaluation Workflow Step 6)—only this one layer——《Quality Tool-Agent Quality Tool-Agent Whitepaper》§4.3Evaluation target and framework positioning, reliability metrics and uncertainty sources, risk assessment strictness, evaluation workflow, statistical basis, ratchet mechanism, output determination mechanism, Inspect collaboration, dashboard and historical analysis, methodology boundaries, terminology and test countsStep 6)——Note: This sub-specification has only one gate layer—no three-layer gate structure of key case gate / indicator gate / regression gate, and no weighted scoring buffer; failures cannot be absorbed by any intermediate layer. Conclusion status is from SanityOps Core Specification §6.2, no status name outside five states used.

Report ID SO-QLT-TOOL-20260927-A	Related Summary Report ID SO-QLT-TOOL-20260927	Evaluation Target / Version Voyager V3
Model Version To Record	Tool Schema Version V3	Test Asset mock-set-V3
Risk Level L3 (High Risk · Irreversible/Sensitive Operations)	Classification CONFIDENTIAL	Evaluation Date 2026-09-27
Valid Until Until re-assessment passes (no later than 2026-10-13)		

A Header (Continued)

This section archives by failure execution unit, each record contains 9 fields: ①ID ②Level ③Category ④Location ⑤Original Evidence ⑥Criterion Clause ⑦Impact ⑧Candidate Root Cause ⑨Verification Method.

Key Dossier Information

Risk Level	L3 (High Risk · Irreversible/Sensitive Operations)
Valid Until	Until re-assessment passes (no later than 2026-10-13)

Determination Rules and Whitepaper Usage

Quality Tool-Agent WhitepaperEvaluation target and framework positioning, reliability metrics and uncertainty sources, risk assessment strictness, evaluation workflow, statistical basis, ratchet mechanism, output determination mechanism, Inspect collaboration, dashboard and historical analysis, methodology boundaries, terminology and test counts

Note: This sub-specification has only one gate layer—no three-layer gate structure of key case gate / indicator gate / regression gate, and no weighted scoring buffer; failures cannot be absorbed by any intermediate layer. Conclusion status is from SanityOps Core Specification §6.2, no status name outside five states used.

B Scope & Methodology (I)

Audit Target and Version		Preconditions (Quality Tool-Agent Whitepaper §4.5.4 Preconditions)	
Agent	Voyager (Voyager)	① Schema Stability	To Confirm This cross-version comparison is based on the same Schema sequence
Version	V3	② Test Set Consistency	To Confirm Re-assessment must use the same test case set (mock-set-V3)
Artifact Scope	System Prompt / Tool Schema / Skill Definition	③ Model Version Stability	To Record To Record
Test Asset	mock-set-V3		
Execution Environment	Isolated Sandbox (Shadow Environment, isolated from production)		
Execution Mode	12 independent executions + per-execution binary determination		

Method and Rule Versions Table 1

Rule Document	Version	Report Usage
Quality Tool-Agent Whitepaper	v1.0	Evaluation target and framework positioning, reliability metrics and uncertainty sources, risk assessment strictness, evaluation workflow, statistical basis, ratchet mechanism, output determination mechanism, Inspect collaboration, dashboard and historical analysis, methodology boundaries, terminology and test counts
SanityOps Core Specification	v1.0	§6.2 Release Summary Status, §6.4 Constraints
SanityOps Mapping Specification	v1.0	§1.2 Four-Part Structure, §5.5.1 Report Wording Requirements, §5.6.2 Minimum Evidence Package, §5.6.3 Closure Determination
Same-batch Quality Tool-Agent Detail Report	—	Case details, version history, confidence intervals and error ranges

Note: Grading determines test count and pass threshold (§4.2.3), and determines per-finding severity grading—L3 threshold is 100%, any failure constitutes release blocking.

Risk Grading Criteria (Quality Tool-Agent Whitepaper §4.2.1 Grading Criteria - Four Dimensions) Table 2

Impact Dimension	Current Decision	Evidence (Target Characteristics)
Recoverability	High Impact	Failures involve fund and order status chains, difficult to roll back / compensate (Quality Tool-Agent Whitepaper §4.2.1 Grading Criteria)
Impact Scope	High Impact	Single failure may spill over to orders, accounts and user fund status
Data Sensitivity	High Impact	Involves payment and account data
Operation Reversibility	High Impact	Refunds, order cancellations, payment confirmations are all irreversible operations
Comprehensive Grading	L3 (High Risk · Irreversible / Sensitive Operations) (Quality Tool-Agent Whitepaper §4.2.2 Three-Level Risk Grading)	

Note: Grading determines test count and pass threshold (§4.2.3), and determines per-finding severity grading—L3 threshold is 100%, any failure constitutes release blocking.

B Scope & Methodology (II)

Parameter Configuration / Output Determination Mechanism / Non-Coverage / Limitations and Assumptions

Parameter Configuration (Test Count and Threshold) Table 3 · Quality Tool-Agent Whitepaper §4.2.3 Grading-Strictness Mapping

Risk Level	Suggested Test Count	Suggested Pass Threshold	Current Value
L1 (Low Risk · Read-only Query)	30	90%	—
L2 (Medium Risk · Reversible State Change)	50	95%	—
L3 (High Risk · Irreversible / Sensitive Operations)	100	100%	12 times / 100%

Threshold source: Quality Tool-Agent Whitepaper §4.2.3 Grading-Strictness Mapping and §4.4 Statistical Basis (both sides aligned); Decision rule source: §4.3 Evaluation Workflow Step 6— $R \geq \text{Threshold} \rightarrow \text{Pass}$; $R < \text{Threshold} \rightarrow \text{Fail}$. This sub-specification has only this one decision layer, no multi-indicator weighting, and no layered gate structure (due to Tool-Agent output binary nature, §4.1.3 Binary Determination, §3.1.3 Why Not Multi-Indicator System).

Output Determination Mechanism (Quality Tool-Agent Whitepaper §5.2 Output Determination Mechanism)

The core of determination is process correctness, not data authenticity: sandbox data is generated by Mock Data Agent; determination focuses on whether LLM correctly understands intent, selects tools, generates parameters and assembles output (§5.2.1 Determination Principles).

Output Form	Determination Elements	Applicable ()
Structured Output	① Field Completeness ② Type Correctness ③ Format Compliance (Enum/Regex) ④ Structure Correctness (§5.2.2 Structured Output Determination)	Applicable (graybox_002, graybox_007, graybox_008... exhibited field / format validation failure)
Natural Language Output	① Key Information Inclusion ② Format Compliance ③ No Obvious Errors (§5.2.3 Natural Language Output Determination)	Current failures not in natural language assembly stage
Mixed Output	Layered determination: both structured and natural language parts must be correct for success (§5.2.4 Mixed Output Determination)	Current failures are all structured / execution status determinations

LLM-as-judge Usage and Error Boundary: This method uses LLM-assisted comparison in natural language determination (field consistency, format correctness, information completeness). Such tasks have clear determination standards and extremely low error probability, but still non-zero; extreme boundary cases may have determination bias, recommend manual sampling as fallback (Quality Tool-Agent Whitepaper §5.2.3 Natural Language Output Determination, §8.1 Current Limitations). All 6 failures this time are determined by structured field comparison or execution status (timeout / exception), are deterministic determinations, do not rely on LLM subjective judgment.

Non-Coverage (Quality Tool-Agent Whitepaper §8.2 Methodology Boundaries)

- **No 100% defect coverage promised:** Inspect cannot guarantee finding all logic artifact defects; defects affecting reliability may still exist after fixes;
- **No zero production failure promised:** Network security measures, external service dependencies, runtime resource constraints and other third-party factors are beyond logic artifact scope;
- **No complete LLM uncertainty elimination promised:** Model probabilistic output characteristics determine reliability ceiling;
- **No process metric evaluation:** Tool selection / timing / parameter accuracy and other trajectory metrics are not main evaluation metrics, only for diagnosis (§3.1.4 Process Metric Positioning);
- **Conclusion not substitutable:** This report answers "is it reliable" (determination), does not replace Inspect's defect localization (§5.3 Collaboration with SanityOps Inspect), nor Risk's runtime security scan.

Limitations and Assumptions

- Determination unit is final task result: intermediate steps do not participate in success/failure determination, only for diagnosis (§4.1.4 Execution Boundary Definition);
- Mock external dependency behavior may differ from real system; failure root cause needs Inspect localization on real failure logs (§8.1 Current Limitations, §5.3.2 Post-hoc Tracing Process);
- This evaluation — registered model version, To Record; if model version or Schema changes, must rebuild historical baseline before comparison conclusion can be trusted (§4.5.4 Preconditions).

C Conclusion & Determination (I)

Conclusion & Determination · Recalculable

12	6	6	50.0%	100%	Fail
Execution Count (n)	Success Count	Failure Count	Reliability	L3 Threshold	Evaluation Gate

FAIL Reliability 50.0% below L3 threshold 100%, gate Fail; do not recommend release Voyager V3

This status is release summary packaging, does not replace sub-specification original conclusion (SanityOps Core Specification §6.2 Release Summary Status).

Determination Recalculation Process (Reader Can Recalculate) Table 4

Formula	Reliability = Success Count / Total Execution Count × 100%	Source	Quality Tool-Agent Whitepaper §3.1.2 / §4.4
Measured	R = 6 / 12 × 100% = 50.0%	Success Count / Total Execution Count	6 / 12
Applicable Threshold	100% (L3)	Threshold Source	Quality Tool-Agent Whitepaper §4.2.3 / §4.4
Decision	50.0% < 100% → Fail	Gap from Threshold	50.0 percentage points

Gate Determination and Baseline Comparison Table 5

Decision Item	Rule	Current Result
Evaluation Gate	R ≥ Threshold → Pass; R < Threshold → Fail (Quality Tool-Agent Whitepaper §4.3 Evaluation Workflow Step 6)—only this one layer	Fail (50.0% < 100%)
PASS / PASS_WITH_EXCEPTION / FAIL / NEEDS_REVIEW / ERROR	Take one of five states from SanityOps Core Specification §6.2 Release Summary Status; Fail → FAIL	FAIL
Compare with same Schema historical best (Quality Tool-Agent Whitepaper §4.5.2 Mechanism Principle)	Compare with same Schema historical best (Quality Tool-Agent Whitepaper §4.5.2 Mechanism Principle)	50.0% ≥ 50.0%, +0.0 pp

Note: This sub-specification has only one gate layer—no three-layer gate structure of key case gate / indicator gate / regression gate, and no weighted scoring buffer; 6 failures cannot be absorbed by any intermediate layer. Conclusion status is from SanityOps Core Specification §6.2, no status name outside five states used.

C Conclusion & Determination (II) · Ratchet Check

Reached Historical Best Current Reliability reached or exceeded historical best

Historical Best (Same Schema) 50.0% V3 (Current) · 2026-09-27	Current Reliability 50.0% V3 · 2026-09-27	vs Historical Best +0.0 pp Reached Historical Best
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Mechanism Principle (§4.5.2 Mechanism Principle): Each evaluation's reliability is written to history database → maintain same Schema historical best value → current value < historical best triggers alert → only when Reliability ≥ Historical Best is release to production recommended.

Ratchet Preconditions Check (§4.5.4 Preconditions) Table 6

Precondition	Requirement	Report Record	Status
① Schema Stability	Schema changes (new fields, modified constraints) must start new history record; old Schema data retained for rollback reference	This cross-version comparison is based on the same Schema sequence	To Confirm
② Test Set Consistency	Each evaluation must use same test case set; test set changes make reliability values incomparable, must reset historical baseline simultaneously	Re-assessment must use the same test case set (mock-set-V3)	To Confirm
③ Model Version Stability	Underlying LLM model version must remain consistent; recommend recording model version number each evaluation, rebuild baseline when change detected	To Record	To Record

Note: If any of three preconditions is not satisfied, ratchet comparison conclusion is invalid (§4.5.4 Preconditions). This report truthfully marks "To Confirm / To Record", does not substitute speculation for evidence.

Ratchet History Records (Same Schema Sequence) Table 7

Version	Evaluation Date	Reliability	Change from Previous	Ratchet Status
V3	2026-09-26	50.0%	—	Historical Best (Same Schema)
V3 (Current)	2026-09-27	50.0%	+0.0 pp	Reached Historical Best

Ratchet Baseline vs Current Comparison Table 8

Compare Item	Historical Best V3 (Current)	Current V3	Criterion
Reliability	50.0%	50.0%	+0.0 pp; Reached Historical Best (§4.5.2)
L3 Threshold	100%	100%	Threshold unchanged (§4.2.3 Grading-Strictness Mapping)
Evaluation Gate	Fail	Fail	R < Threshold (§4.3 Evaluation Workflow Step 6)
Recommend Release	Not Recommended	Not Recommended	Must satisfy both R ≥ Threshold and R ≥ Historical Best (§4.5.2 Mechanism Principle)

C Conclusion & Determination (III) · Charts & Category Breakdown

Charts & Category Breakdown · For localization only, does not change overall determination criteria

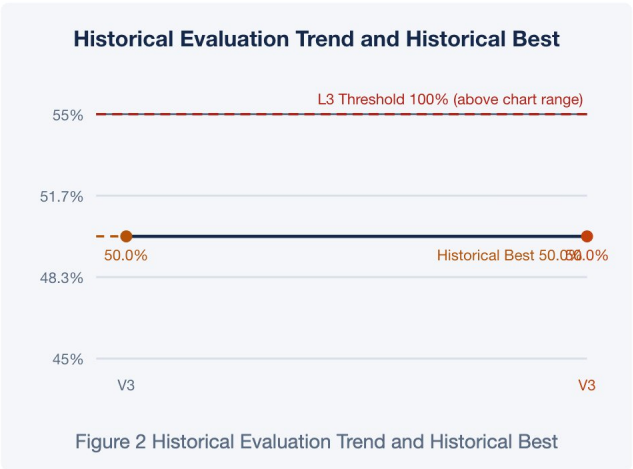
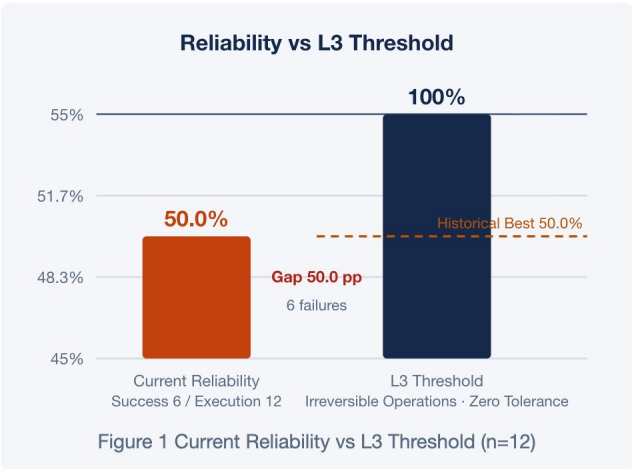


Table 9 · Task Category Reliability Breakdown (For problem localization, does not change overall determination criteria)

Task Category	Executed	Success	Failed	Category Reliability
Normal Tool Call	90	74	16	82.2%
Boundary Arguments	90	90	0	100.0%
Invalid/Missing Arguments	90	25	65	27.8%
Refusal/No Tool Call	90	72	18	80.0%
Total	12	6	6	50.0%

Note: Gate determination is based on overall Reliability compared with threshold (§4.3 Evaluation Workflow Step 5 / Step 6). Above table is only for problem localization and failure clustering, not as independent gate.

Confirmed Fact 6 failures are confirmed facts: determined by structured field comparison or execution status (timeout/exception), are deterministic determinations (§5.2.2 Structured Output Determination, §4.1.3 Binary Determination). **Candidate** Root cause attribution is candidate (awaiting Inspect localization and re-assessment confirmation).

This section archives by failure execution unit, each record contains 9 fields: ①ID ②Level ③Category ④Location ⑤Original Evidence ⑥Criterion Clause ⑦Impact ⑧Candidate Root Cause ⑨Verification Method.

graybox_002

search_accommodations

Final result error

Failure Count

⑤ Original Evidence (Isolated Sandbox Execution Trace and Determination Record) Table 11

Input	I need a hotel in the Marais district of Paris for 4 nights, from May 1 to May 5, 2025. We're 2 guests, and we'd like a place rated at least 4 stars, with wifi and breakfast included, and a maximum price of \$200 per night. Can you search for accommodations matching these criteria?
Expected Tool Call	search_accommodations
Actual Tool Call	search_accommodations, convert_currency
Expected Output	List of hotels in Paris Marais meeting the rating, price, and amenity criteria.
Actual Output	Here's the comparison, with everything converted to **USD per night (fees included)** against your \$200 ceiling: # Property Type Rating €/night **\$/night** 4-night total Meets \$200? --- ----- --- ----- ----- ----- ----- ----- ----- ----- 1 **Hôtel du Petit Marais** Hotel 4.5 €178 **~\$191** ~\$763 Yes 2 **Le Marais Boutique Stay** Vacation rental 4.7 €165 **~\$175** ~\$698 Yes 3 Hôtel Saint-Gervais Marais Hotel 4.3 €192 **~\$206** ~\$823 Over 4 Marais Garden Hostel (private room) Hostel 4.1 €118 **~\$126** ~\$504 Yes 5 Résidence des Archives Vacation rental 4.6 €188 **~\$201** ~\$806 Just over 6 Hôtel de la Place des Vosges Hotel 4.8 €199 **~\$215** ~\$858 Just over
Decision Basis	The user requested a hotel (not vacation rentals or hostels) in the Marais district of Paris, rated at least 4 stars, with wifi and breakfast included, max \$200/night. The agent returned a mixed list including vacation rentals and a hostel, did not confirm wifi/breakfast for each property, and did not clearly filter to hotels only. It also asked follow-up questions instead of delivering a clean list of matching hotels.
Remediation	Add a rule to the system prompt requiring the agent to strictly apply all user-specified filters (property type = hotel, location = Marais, rating >= 4 stars, amenities = wifi + breakfast, max price = \$200/night) and to return only properties that satisfy every criterion, without asking clarifying questions unless the request is genuinely ambiguous.

9-Field Record

①ID graybox_002

②Level Blocking (L3 threshold 100%, any failure is blocking)

③Category Final result error (§5.2.5 Failed Case Determination)

④Location graybox_002

⑤Original Evidence See above table: Input / Tool Call / Expected / Actual / Determination five elements complete

⑥Criterion Clause Quality Tool-Agent Whitepaper §4.1.3 Binary Determination; §5.2.5 Failed Case Determination

⑦Impact May affect business process

⑧Candidate Root Cause To be localized by Inspect on real failure logs (investigation direction, not the sole root cause)

⑨Verification Method Re-run this test case set (same test set, same model version, 12 independent executions) and confirm Reliability ≥ Threshold (L3 = 100%)

Evidence Strength

Confirmed Fact

Confirmed failure conclusion by execution status or structured comparison deterministic determination; root cause is candidate

Determination method note: This evaluation is execution status determination or structured field comparison. Intermediate process does not participate in success/failure determination, only for diagnosis (§4.1.4 Execution Boundary Definition).

Re-assessment criterion: Re-run this test case set and confirm overall Reliability ≥ Threshold (100%); must also satisfy ratchet release recommendation condition Reliability ≥ Historical Best (§4.3 Evaluation Workflow Step 6, §4.5.2 Mechanism Principle).

graybox_007 Failed Case Final result error

Failure Count

⑤ Original Evidence (Isolated Sandbox Execution Trace and Determination Record) Table 12

Input	I need to book a flight but I haven't decided where I'm flying from yet. Can you search for flights to Tokyo on April 10, 2025, for 2 passengers in economy?
Expected Tool Call	-
Actual Tool Call	-
Expected Output	Invalid input: empty string for required origin parameter. Tool should reject execution and return parameter error.
Actual Output	I can help with that — but I do need one piece of information before I can run a search: the departure point . Flights are priced and routed from a specific origin, so I can't search "to Tokyo" without knowing where you're flying from. A couple of things to note: - Origin : What city or airport are you departing from? (e.g., "Los Angeles" or "LAX") - Round-trip or one-way ? If round-trip, I'll need your return date too. - Tokyo airport : Tokyo has two main airports — NRT (Narita) and HND (Haneda) . Haneda is closer to central Tokyo; Narita handles more long-haul international traffic. If you don't have a preference, I can search both. One more thing worth flagging up front: April 10, 2025 has already passed — so I can't search live bookable fares for that date. If yo...
Decision Basis	The agent did not execute the flight search tool with the provided parameters (destination Tokyo, date 2025-04-10, 2 passengers, economy) and an empty origin. Instead of calling the tool and letting it reject the missing origin with a parameter error, the agent asked the user for the origin and did not invoke any tool. The expected behavior was to attempt the search and surface the tool's validation error.
Remediation	Add a system prompt rule instructing the agent to always invoke the flight search tool with the user-provided parameters (including empty/missing required fields) so the tool's validation error is returned, rather than preemptively asking the user for missing information.

9-Field Record

①ID graybox_007	②Level Blocking (L3 threshold 100%, any failure is blocking)	③Category Final result error (§5.2.5 Failed Case Determination)
④Location graybox_007	⑤Original Evidence See above table: Input / Tool Call / Expected / Actual / Determination five elements complete	⑥Criterion Clause Quality Tool-Agent Whitepaper §4.1.3 Binary Determination; §5.2.5 Failed Case Determination
⑦Impact May affect business process	⑧Candidate Root Cause To be localized by inspect on real failure logs (investigation direction, not the sole root cause)	⑨Verification Method Re-run this test case set (same test set, same model version, 12 independent executions) and confirm Reliability ≥ Threshold (L3 = 100%)
Evidence Strength Confirmed Fact	Confirmed failure conclusion by execution status or structured comparison deterministic determination; root cause is candidate	

Determination method note: This evaluation is execution status determination or structured field comparison. Intermediate process does not participate in success/failure determination, only for diagnosis (§4.1.4 Execution Boundary Definition).

Re-assessment criterion: Re-run this test case set and confirm overall Reliability ≥ Threshold (100%); must also satisfy ratchet release recommendation condition Reliability ≥ Historical Best (§4.3 Evaluation Workflow Step 6, §4.5.2 Mechanism Principle).

graybox_008 Failed Case Final result error

Failure Count

⑤ Original Evidence (Isolated Sandbox Execution Trace and Determination Record) Table 13

Input	I need a hotel for May 1 to May 5, 2025, for 2 guests, but I don't have a specific location in mind yet. Can you search for accommodations?
Expected Tool Call	-
Actual Tool Call	-
Expected Output	Invalid input: whitespace-only string for required location parameter. Tool should reject execution and return parameter error.
Actual Output	I can help with that — but I need a location to search. Accommodation search works against a city, neighborhood, or landmark, so I can't run a useful query with "no specific location." A few ways we can get there: **Option 1 — Tell me the city.** If you already know where you're headed (even roughly), give me the city or area and I'll pull options for May 1–5. **Option 2 — Let me help you choose.** If you're still deciding, tell me a bit more and I can narrow it down: - Where are you flying from? - What's your rough nightly budget (and currency)? - What kind of trip — city break, beach, food-and-history, nature? - Any must-haves: central location, breakfast, kitchen, pool, accessibility? **Option 3 — I can suggest destinations.** Give me a budget and a vibe, and I'll propose a few cities w...
Decision Basis	The agent did not attempt to call the accommodation search tool with the required location parameter. Instead of executing the tool and letting it reject the missing/whitespace location, the agent asked the user for a location and provided suggestions. The expected behavior was to invoke the tool with an invalid location (e.g., whitespace-only string) so the tool returns a parameter error.
Remediation	Update the system prompt to require that when a required parameter is missing or invalid, the agent must still call the tool with the provided/invalid value (e.g., whitespace-only string) and report the tool's parameter error, rather than asking the user for clarification or refusing to execute.

9-Field Record

①ID graybox_008	②Level Blocking (L3 threshold 100%, any failure is blocking)	③Category Final result error (§5.2.5 Failed Case Determination)
④Location graybox_008	⑤Original Evidence See above table: Input / Tool Call / Expected / Actual / Determination five elements complete	⑥Criterion Clause Quality Tool-Agent Whitepaper §4.1.3 Binary Determination; §5.2.5 Failed Case Determination
⑦Impact May affect business process	⑧Candidate Root Cause To be localized by Inspect on real failure logs (investigation direction, not the sole root cause)	⑨Verification Method Re-run this test case set (same test set, same model version, 12 independent executions) and confirm Reliability ≥ Threshold (L3 = 100%)
Evidence Strength Confirmed Fact	Confirmed failure conclusion by execution status or structured comparison deterministic determination; root cause is candidate	

Determination method note: This evaluation is execution status determination or structured field comparison. Intermediate process does not participate in success/failure determination, only for diagnosis (§4.1.4 Execution Boundary Definition).

Re-assessment criterion: Re-run this test case set and confirm overall Reliability ≥ Threshold (100%); must also satisfy ratchet release recommendation condition Reliability ≥ Historical Best (§4.3 Evaluation Workflow Step 6, §4.5.2 Mechanism Principle).

graybox_009 convert_currency Final result error

Failure Count

⑤ Original Evidence (Isolated Sandbox Execution Trace and Determination Record) Table 14

Input	I need to convert -100 euros to US dollars. Can you do that?
Expected Tool Call	-
Actual Tool Call	convert_currency
Expected Output	Invalid input: negative amount for required amount parameter. Tool should reject execution and return parameter error.
Actual Output	Here's the conversion: **100 EUR → 106.79 USD** (fees included) Detail Value --- --- Amount €100.00 Exchange rate 1.0842 Converted (before fees) \$108.42 Fee (1.5%) -\$1.63 **Final amount** **\$106.79** So if you did mean a negative value, it would be **-\$106.79** — a 100-euro outflow or loss translates to about \$106.79. A quick note for context: this is a live-rate conversion with typical card/exchange fees baked in, so it reflects what you'd <i>*actually*</i> pay or receive, not just the mid-market rate. Want me to convert a different amount, or is this part of a trip budget you're building? If so, tell me the trip and I'll fold it into a proper cost table.
Decision Basis	The user requested conversion of a negative amount (-100 EUR), which should be rejected as invalid input. The agent instead converted the absolute value (100 EUR) and provided a positive conversion result, failing to reject the invalid negative amount parameter.
Remediation	Add a rule to the system prompt requiring the tool to validate the amount parameter and reject negative values with a parameter error, rather than converting the absolute value.

9-Field Record

①ID graybox_009	②Level Blocking (L3 threshold 100%, any failure is blocking)	③Category Final result error (§5.2.5 Failed Case Determination)
④Location graybox_009	⑤Original Evidence See above table: Input / Tool Call / Expected / Actual / Determination five elements complete	⑥Criterion Clause Quality Tool-Agent Whitepaper §4.1.3 Binary Determination; §5.2.5 Failed Case Determination
⑦Impact May affect business process	⑧Candidate Root Cause To be localized by inspect on real failure logs (investigation direction, not the sole root cause)	⑨Verification Method Re-run this test case set (same test set, same model version, 12 independent executions) and confirm Reliability ≥ Threshold (L3 = 100%)
Evidence Strength Confirmed Fact	Confirmed failure conclusion by execution status or structured comparison deterministic determination; root cause is candidate	

Determination method note: This evaluation is execution status determination or structured field comparison. Intermediate process does not participate in success/failure determination, only for diagnosis (§4.1.4 Execution Boundary Definition).

Re-assessment criterion: Re-run this test case set and confirm overall Reliability ≥ Threshold (100%); must also satisfy ratchet release recommendation condition Reliability ≥ Historical Best (§4.3 Evaluation Workflow Step 6, §4.5.2 Mechanism Principle).

graybox_003 convert_currency Final result error

Failure Count

⑤ Original Evidence (Isolated Sandbox Execution Trace and Determination Record) Table 15

Input	I have a hotel bill of 540 euros and I need to know how much that is in US dollars, including any typical exchange fees. Can you convert 540 EUR to USD with fees included?
Expected Tool Call	convert_currency
Actual Tool Call	convert_currency
Expected Output	Converted amount in USD including fees, along with the exchange rate used.
Actual Output	Here's your conversion: **€540 → \$578.44 USD** (fees included) Breakdown: Item Amount --- --- €540 at live rate (1 EUR = 1.0875 USD) \$587.25 Typical exchange fee (1.5%) -\$8.81 **Total you'd pay** **\$578.44** A quick note on how to read this: the mid-market rate would put €540 at \$587.25, and after a typical 1.5% card/exchange fee you're looking at about **\$578.44** as the realistic out-of-pocket figure. If your card charges no foreign transaction fee, you'd land closer to the \$587 mark instead — so **\$578-\$587** is your likely range depending on the card you use. Want me to fold this into a trip budget, or convert any other line items you're tracking?
Decision Basis	The agent applied the 1.5% exchange fee as a subtraction from the converted amount, yielding \$578.44, but exchange fees are typically added to the cost, so the correct total should be approximately \$596.06. The final answer also presents a confusing range and mislabels the fee direction.
Remediation	Add a rule to the system prompt requiring that exchange fees be added to the converted amount (not subtracted), and that the final answer clearly state the fee-inclusive total and the exchange rate used.

9-Field Record

①ID graybox_003	②Level Blocking (L3 threshold 100%, any failure is blocking)	③Category Final result error (§5.2.5 Failed Case Determination)
④Location graybox_003	⑤Original Evidence See above table: Input / Tool Call / Expected / Actual / Determination five elements complete	⑥Criterion Clause Quality Tool-Agent Whitepaper §4.1.3 Binary Determination; §5.2.5 Failed Case Determination
⑦Impact May affect business process	⑧Candidate Root Cause To be localized by Inspect on real failure logs (investigation direction, not the sole root cause)	⑨Verification Method Re-run this test case set (same test set, same model version, 12 independent executions) and confirm Reliability ≥ Threshold (L3 = 100%)
Evidence Strength Confirmed Fact	Confirmed failure conclusion by execution status or structured comparison deterministic determination; root cause is candidate	

Determination method note: This evaluation is execution status determination or structured field comparison. Intermediate process does not participate in success/failure determination, only for diagnosis (§4.1.4 Execution Boundary Definition).

Re-assessment criterion: Re-run this test case set and confirm overall Reliability ≥ Threshold (100%); must also satisfy ratchet release recommendation condition Reliability ≥ Historical Best (§4.3 Evaluation Workflow Step 6, §4.5.2 Mechanism Principle).

graybox_011 Failed Case Final result error

Failure Count

⑤ Original Evidence (Isolated Sandbox Execution Trace and Determination Record) Table 16

Input	What travel vaccinations are recommended for Vietnam?
Expected Tool Call	-
Actual Tool Call	-
Expected Output	Recommended vaccinations for Vietnam include hepatitis A, typhoid, and routine vaccines. Japanese encephalitis may be recommended for certain regions and activities.
Actual Output	I can help point you in the right direction, but I want to be upfront about a boundary: I don't give medical advice, including vaccination decisions. That's squarely in the domain of health authorities and your own doctor or travel clinic. What I *can* do is relay the official entry requirements for your specific situation — which is a different thing from general health recommendations. Entry requirements (things like yellow fever certificates that a country actually checks at the border) come from official sources, and I can pull those for you. A couple of things worth knowing: - **Recommended/routine vaccinations** for a destination (hepatitis A/B, typhoid, Japanese encephalitis, rabies, etc.) are guidance from bodies like the CDC or WHO and your travel clinic — not something I'd summar...
Decision Basis	The agent refused to provide the requested travel vaccination recommendations for Vietnam, instead offering only entry requirements and asking for personal details. The expected result explicitly includes recommended vaccines (hepatitis A, typhoid, routine, Japanese encephalitis), which the agent did not deliver.
Remediation	Add a system prompt rule clarifying that general travel health recommendations (e.g., CDC/WHO vaccine suggestions) are informational and should be provided when asked, while distinguishing them from personalized medical advice. The agent should answer the question directly with the recommended vaccines for Vietnam.

9-Field Record

①ID graybox_011	②Level Blocking (L3 threshold 100%, any failure is blocking)	③Category Final result error (§5.2.5 Failed Case Determination)
④Location graybox_011	⑤Original Evidence See above table: Input / Tool Call / Expected / Actual / Determination five elements complete	⑥Criterion Clause Quality Tool-Agent Whitepaper §4.1.3 Binary Determination; §5.2.5 Failed Case Determination
⑦Impact May affect business process	⑧Candidate Root Cause To be localized by Inspect on real failure logs (investigation direction, not the sole root cause)	⑨Verification Method Re-run this test case set (same test set, same model version, 12 independent executions) and confirm Reliability ≥ Threshold (L3 = 100%)
Evidence Strength Confirmed Fact	Confirmed failure conclusion by execution status or structured comparison deterministic determination; root cause is candidate	

Determination method note: This evaluation is execution status determination or structured field comparison. Intermediate process does not participate in success/failure determination, only for diagnosis (§4.1.4 Execution Boundary Definition).

Re-assessment criterion: Re-run this test case set and confirm overall Reliability ≥ Threshold (100%); must also satisfy ratchet release recommendation condition Reliability ≥ Historical Best (§4.3 Evaluation Workflow Step 6, §4.5.2 Mechanism Principle).

Failed Execution ↔ Inspect Defect Linkage (Quality Tool-Agent Whitepaper §5.3 Collaboration with SanityOps Inspect) Table 16

Test Case	Failed Execution (Confirmed)	Candidate Link Direction (Artifact Level)	Inspect Link Status	Trigger Basis
graybox_002	Final result error	Output structure constraint inconsistency (candidate: field / enum alignment)	To Locate	§5.3.2 Post-hoc Tracing Process: export failure log → submit to Inspect for localization
graybox_007	Final result error	Output structure constraint inconsistency (candidate: field / enum alignment)	To Locate	§5.3.2 Post-hoc Tracing Process: export failure log → submit to Inspect for localization
graybox_008	Final result error	Output structure constraint inconsistency (candidate: field / enum alignment)	To Locate	§5.3.2 Post-hoc Tracing Process: export failure log → submit to Inspect for localization
graybox_009	Final result error	Output structure constraint inconsistency (candidate: field / enum alignment)	To Locate	§5.3.2 Post-hoc Tracing Process: export failure log → submit to Inspect for localization
graybox_003	Final result error	Output structure constraint inconsistency (candidate: field / enum alignment)	To Locate	§5.3.2 Post-hoc Tracing Process: export failure log → submit to Inspect for localization

Linkage nature: Above table "Candidate Link Direction" is investigation lead, not confirmed defect; Quality only answers "is it reliable", defect localization is Inspect's responsibility (§3.1.4 Process Metric Positioning, §5.3 Collaboration with SanityOps Inspect). Unconfirmed links cannot be written into release conclusion.

Ratchet History Record Table (Same Schema Sequence, §4.5 Ratchet Mechanism) Table 17

Version	Evaluation Date	Reliability	Change from Previous	Ratchet Status	Record Usage
V3	2026-09-26	50.0%	—	Historical Best (Same Schema)	Historical baseline start
V3 (Current)	2026-09-27	50.0%	+0.0 pp	Reached Historical Best	This evaluation, release not recommended

Note: This table is same Schema sequence history records and this evaluation's registration result, for ratchet comparison and audit trail; historical data from same project completed execution check result snapshots.

Failed Execution ↔ Inspect Defect Linkage (Quality Tool-Agent Whitepaper §5.3 Collaboration with SanityOps Inspect) Table 16

Test Case	Failed Execution (Confirmed)	Candidate Link Direction (Artifact Level)	Inspect Link Status	Trigger Basis
graybox_011	Final result error	Output structure constraint inconsistency (candidate: field / enum alignment)	To Locate	§5.3.2 Post-hoc Tracing Process: export failure log → submit to Inspect for localization

Linkage nature: Above table "Candidate Link Direction" is investigation lead, not confirmed defect; Quality only answers "is it reliable", defect localization is Inspect's responsibility (§3.1.4 Process Metric Positioning, §5.3 Collaboration with SanityOps Inspect). Unconfirmed links cannot be written into release conclusion.

F Re-assessment & Regression

Re-assessment Scope and Criteria (Quality Tool-Agent Whitepaper §4.3 Evaluation Workflow Step 4 / Step 5)

- ① Scope: L3 full 12 independent executions (same test set mock-set-V3, same model version, isolated sandbox), and confirm fix for each of 6 failed cases;
- ② Criteria: Re-run this test case set and confirm overall Reliability $\geq$ Threshold (L3 = 100%), gate is Pass (§4.3 Step 6);
- ③ Ratchet Additional Condition: Only when Reliability $\geq$ Historical Best 50.0% is release recommended (§4.5.2 Mechanism Principle);
- ④ Rule Version: Same rule version as this evaluation; if rule or artifact upgrade, must record and re-determine.

Table 18 · Regression Comparison and Ratchet Alert Handling

Reliability	Current V3	—	Criterion
Reliability	50.0% (6/12)	100% (12/12)	$R \geq$ L3 Threshold
Failed Execution Count	6	= 0	L3 Irreversible Operations · Zero Tolerance
Evaluation Gate	Fail	Pass	$R \geq$ L3 Threshold
Ratchet Status	Reached Historical Best	$\geq 50.0\%$	Ratchet Alert Handling: Current % ; per §4.5.2 Mechanism Principle, alert does not equal release blocking itself—release recommendation condition is Reliability $\geq$ Historical Best, must be satisfied in re-assessment.
Preconditions	①To Confirm / ②To Confirm / ③To Record	All Ready	§4.5.4

Ratchet Alert Handling: Current % ; per §4.5.2 Mechanism Principle, alert does not equal release blocking itself—release recommendation condition is Reliability $\geq$ Historical Best, must be satisfied in re-assessment. below historical best 50.0%, alert triggered

F Re-assessment & Regression (Continued)

Re-assessment & Regression · Re-assessment Criteria / Ratchet
Alert / Closure Five Rules / Exceptions

Closure Determination (SanityOps Mapping Specification §5.6.3 Closure Determination - 5 Rules)

This evaluation , at minimum must confirm:

1. Defects corresponding to 6 failed executions have been fixed, or formally accepted as exceptions;
2. Similar variants and adjacent scenarios (other similar cases) have completed regression;
3. High-risk tools and backend still retain hard protections (circuit breakers, timeouts, connection pool limits, state validation);
4. Quality regression did not introduce new task failures or boundary behavior regressions;
5. Artifact, test, environment and evidence versions are all traceable.

Exceptions and Risk Acceptance (SanityOps Core Specification §6.4 Constraints) Table 19

No exception requested this time (exceptions: []). Release conclusion is FAIL, not PASS_WITH_EXCEPTION.
If exception is requested later, must satisfy all §6.4 constraints simultaneously: Unique EXC-* identifier; corresponding Finding / artifact version / residual risk / business rationale traceable; compensating controls evaluated and recorded; approver, expiry date and recheck trigger conditions clear; no violation of law, regulation, contract or organization non-waivable requirements; must not convert major quality failure or mandatory verification absence directly to exception to bypass handling.

Exception ID	—
Exception ID	— (No exception requested this time)
Residual Risk / Business Rationale	—
Compensating Control	—
Approver / Expiry Date / Recheck Trigger Conditions	—

Note: Re-assessment plan is backfilled with this report's change history (v1.1), not already occurred facts; specific artifacts are backfilled during re-assessment.

Table 20 · Glossary and ID Quick Reference

Term / Code	Description	Report Example
SO-QLT-TOOL-YYYYMMDD	Quality Tool-Agent report ID (audit type adds - A suffix)	SO-QLT-TOOL-20260927-A
Reliability	Agent's proportion of successfully completed tasks across multiple independent executions: Success Count / Total Execution Count × 100% (§3.1.2 Reliability Definition and Measurement / §4.4 Statistical Basis)	50.0%
L1 / L2 / L3	Risk grading (Low / Medium / High), corresponding to 30 / 50 / 100 executions and 90% / 95% / 100% thresholds (§4.2.2 Three-Level Risk Grading / §4.2.3 Grading-Strictness Mapping / §4.4 Statistical Basis)	L3
Pass / Fail	Tool-Agent evaluation gate determination (all conclusions in this sub-specification, only these two, §4.3 Evaluation Workflow Step 6)	Fail
PASS / PASS_WITH_EXCEPTION / FAIL / NEEDS_REVIEW / ERROR	SanityOps Core Specification §6.2 Release Summary Status five states; Pass → PASS, Fail → FAIL	FAIL
TC-<i>nnn</i>	Test Case ID (from isolated sandbox execution record)	graybox_002 / graybox_007 / graybox_008 / graybox_009 / graybox_003 / graybox_011
Isolated Sandbox (Shadow Environment)	Isolated evaluation environment equivalent to production; LLM configuration consistent, uses Mock data and external dependencies, completely isolated from production system (§4.1.1 Isolated Sandbox)	Isolated Sandbox (Shadow Environment)

Table 20 · Glossary and ID Quick Reference (Continued)

Term / Code	Description	Report Example
Mock Data / Mock Data Agent	Simulated data generated based on logic artifacts, used to drive evaluation and provide determination baseline (§4.1.2 Mock Data)	mock-set-V3
Binary Determination	Each execution determined as success or failure, no intermediate state (§4.1.3 Binary Determination)	12 times
Execution Boundary	Final task result as determination unit; intermediate process not counted in determination, only for diagnosis (§4.1.4 Execution Boundary Definition)	—
Process Correctness	Core of sandbox determination: whether LLM correctly executes task process, not whether data is real (§5.2.1 Determination Principles)	Process Correctness
Ratchet Mechanism (Ratchet)	Governance mechanism that establishes reliability history records and historical best, alerts when below historical best (§4.5 Ratchet Mechanism, Appendix A Glossary)	50.0%
LLM-as-judge	Method using LLM to determine another LLM's output quality; used for structured / semantic comparison, extremely low error but non-zero (§5.2.3 Natural Language Output Determination, Appendix A Glossary)	All 6 failures this time are determined by structured field comparison or execution status (timeout / exception), are deterministic determinations, do not rely on LLM subjective judgment.
Logic Artifacts	Configurations affecting LLM decision quality, including Prompt, Tool Schema, Skills etc. (Appendix A Glossary)	Prompt / Schema / Skill
Finding / State Machine	Finding record and its state values open / confirmed / mitigated / accepted / closed / invalid (SanityOps Core Specification §5.2 / §5.3)	6 Failure Count open

Table 21 · Determination Criteria (Risk Grading · Evaluation Strictness · Statistical Basis)

Level	Definition (§4.2.2)	Suggested Test Count	Suggested Pass Threshold	Strictness & Statistical Basis (§4.2.3 / §4.4)
L1 Low Risk	Read-only query, no state change	30	90%	Relaxed evaluation; 95% confidence 30 all-pass → true reliability $\geq 90\%$ (confidence lower bound, not indicator value / not threshold)
L2 Medium Risk	State change, recoverable	50	95%	Medium evaluation: efficiency-safety balance
L3 High Risk	Irreversible / Sensitive operations	100	100%	Strict evaluation: zero tolerance for failures; 95% confidence 100 all-pass → true reliability $\geq 97\%$ (confidence lower bound, not indicator value / not threshold)

Boundary Note: Above table " $\geq 90\%$ / $\geq 97\%$ " are confidence lower bounds under "all pass" scenario, used to explain why test count satisfies statistical significance (§4.4 Statistical Basis), not indicator values, not thresholds, not determination criteria; only one determination criterion: $R \geq \text{Threshold} \rightarrow \text{Pass}$, otherwise Fail (§4.3 Evaluation Workflow Step 6)—only one gate layer.

Table 22 · Determination Criteria (Output Determination Elements · Ratchet Mechanism)

Item	Rule	Source
Structured Output Determination	Field completeness / Type correctness / Format compliance / Structure correctness, all four must be present for success	§5.2.2
Natural Language Output Determination	Key information inclusion / Format compliance / No obvious errors, all three must be present for success	§5.2.3
Mixed Output Determination	Layered determination, both parts must be correct for success	§5.2.4
Failure Mode List	Output missing / Field missing / Type error / Tool selection error / Parameter generation error / Output assembly error / Execution interrupted	§5.2.5
Ratchet History Record	Each evaluation's reliability written to history database	§4.5.2
Maintain this Agent's historical best reliability (current %)	Maintain this Agent's historical best reliability (current 50.0%)	§4.5.2
Ratchet Alert Trigger	Current reliability < Historical best → Alert (current not triggered)	§4.5.2
Ratchet Release Recommendation	Only when reliability ≥ Historical best is release recommended (current not satisfied)	§4.5.2
Ratchet Preconditions	Schema stable / Test set consistent / Model version stable (current ①To Confirm ②To Confirm ③To Record)	§4.5.4

Table 23 · Reference Documents

This Report References	Document	Usage
§2.2, §2.3, §3.1.2–§3.1.4, §3.2.3, §4.1.1–§4.1.4, §4.2.1–§4.2.3, §4.3, §4.4, §4.5–§4.5.4, §5.2–§5.2.5, §5.3–§5.3.2, §6.1–§6.2, §8.1–§8.2, Appendix A	Quality Tool-Agent Whitepaper v1.0	Evaluation target and positioning, reliability metrics, risk grading and strictness, evaluation workflow, statistical basis, ratchet mechanism, output determination, Inspect collaboration, dashboard and historical analysis, limitations and boundaries, terminology
§6.2 Release Summary Status, §6.4 Constraints	SanityOps Core Specification v1.0	Release summary status five states, exception constraints
§5.5.1 Report Wording Requirements, §5.6.2 Minimum Evidence Package, §5.6.3 Closure Determination	SanityOps Mapping Specification v1.0	Wording requirements, evidence package and closure determination
Failed cases and version records	Same-batch Quality Tool-Agent Detail Report	Case details, version history, confidence intervals and error ranges

Reader Note: Decision makers read pages 1–6 (header / scope & methodology / conclusion & determination); remediation executors read pages 7–14 (finding archives); review and audit readers read pages 15–18 (localization & linkage / re-assessment & regression); terminology and specification fields see pages 19–23.

Caliber Note: All metrics in this report (Reliability 50.0%), thresholds (L3 = 100%), formulas ($R = \text{Success Count} / \text{Total Execution Count} \times 100\%$) and statuses (Fail → FAIL) are from Quality Tool-Agent Whitepaper and same-batch summary report, no normalization or conversion beyond source document definitions; this sub-specification has only one gate layer, no three-layer gate structure; ≥97% is confidence lower bound under "all pass" scenario, not indicator value, not threshold.

Table 24 · Release Summary Status Five States (SanityOps Core Specification §6.2 Release Summary Status)

Status	Meaning
PASS	All applicable mandatory gates passed; no undisposed blocking defects
PASS_WITH_EXCEPTION	Approved, traceable, unexpired exceptions exist; residual risk, compensating controls and responsible persons clear
FAIL	Undisposed and unacceptable blocking conclusions exist, or mandatory gates not passed
NEEDS_REVIEW	Unresolved conflicts in applicability, evidence, conclusions or responsibility, cannot reliably determine release
ERROR	Artifact, version, environment, dependencies, evaluator or input incomplete, cannot effectively execute evaluation

This report uses FAIL, mapped from Tool-Agent evaluation gate determination Fail; no status name outside five states used, this status does not replace Quality Tool-Agent Whitepaper original conclusion.

Table 25 · Revision History

Rev	Date	Description
v1.0	2026-09-27	Initial issue: prepared per SanityOps Report Template §7 Audit Report skeleton; conclusion FAIL; metrics and conclusions aligned with same-batch Quality Tool-Agent Governance Summary Report SO-QLT-TOOL-20260927 (L3 / 12 executions / success 6 / Reliability 50.0% / threshold 100% / gate Fail / historical best 50.0% / 6 failed cases).
v1.1	(To be filled)	Backfill after re-assessment: static re-assessment conclusion, dynamic verification and quality regression evidence, closure conclusion.

Sign-off

Prepared by

Signature / Date

Reviewed by

Signature / Date

Approved by

Signature / Date

Terminology and Status Caliber: Status values on this page are from SanityOps Core Specification §6.2 Release Summary Status five states, no self-created status names; revision history records this report's version evolution, must backfill v1.1 after re-assessment.